

Exp No: 15 EMBEDDED C PROGRAM TO CONVERT THE HEXADECIMAL DATA 0XCFH TO DECIMAL AND DISPLAY THE DIGITS ON PORTS P0, P1 AND P2

Github link: <https://github.com/Pardhav2006/embedded-system>

PROGRAM:

```
#include <reg51.h>

void main(void)
{
    unsigned char hexa = 0x08;
    unsigned char hundreds, tens, units;
    unsigned char decimal;
    decimal = hexa;    // Hex 08H = Decimal 8
    hundreds = decimal / 100;
    decimal = decimal % 100;
    tens    = decimal / 10;
    units   = decimal % 10;
    P0 = units;    // Output units digit
    P1 = tens;     // Output tens digit
    P2 = hundreds; // Output hundreds digit
    while(1);
}
```

CAUsers\yadaf\OneDrive\Documents\embedded\exp 15\proj - uVision

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Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
a	0x00
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0000
PC	0x0000
status	0x0000
sec	0x0004
psw	0x00

Disassembly

```
21: while(1);  
C:\0x0821 80FE JMP C:\0821  
C:\0x0823 787F MOV R0, #0x7F  
C:\0x0825 E4 CLR A
```

exp_15.c

```
1 #include <reg51.h>  
2  
3 void main(void)  
4 {  
5     unsigned char hexa = 0x08;  
6     unsigned char hundreds, tens, units;  
7     unsigned char decimal;  
8  
9     decimal = hexa; // Hex 08H = Decimal 8  
10  
11     hundreds = decimal / 100;  
12     decimal = decimal % 100;  
13  
14     tens = decimal / 10;  
15     units = decimal % 10;  
16  
17     P0 = units; // Output units digit  
18     P1 = tens; // Output tens digit  
19     P2 = hundreds; // Output hundreds digit  
20  
21     while(1);  
22 }
```

Parallel Port 0

Port 0: 7 Bits 0
P0: 0x08
Pins: 0x08

Parallel Port 1

Port 1: 7 Bits 0
P1: 0x00
Pins: 0x00

Parallel Port 2

Port 2: 7 Bits 0
P2: 0x00
Pins: 0x00

Command

Running with Code Size Limit: 2K
Load "C:\Users\yadaf\OneDrive\Documents\embedded\Objects\exp 15"

Watch 1

Name	Value	Type
main	C:\0800	function
hexa	0x08	uchar
hundreds	0x00	uchar
tens	0x00	uchar
units	0x08	uchar

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE

Simulation

11: 127.48486328 sec IL21 C14 CAP: NUM SCRL OVR: R/W